AI & ML Internship Program

Task 6: Social Media Attention Span Study

Domain: Statistics + Behavioral Analytics + Data Analytics
Difficulty: Beginner → Intermediate
Industry Focus: Social Media / User Analytics / Product Intelligence / Consumer Behavior
Skills Covered: Statistical Analysis, EDA, Hypothesis Testing, Correlation Analysis, User Behavior Analytics, Data Storytelling

Business Problem

Social media platforms compete for one valuable resource: **user attention**.

Understanding user attention span is critical because it directly affects:

- User engagement
- Content recommendations
- Ad revenue
- Session retention
- Platform growth

Product teams want answers:

- How long do users stay engaged?
- What causes users to lose attention?
- Does content type affect session duration?
- Do age groups consume content differently?
- Does excessive scrolling reduce engagement?

As a Data Analyst, your task is to investigate user behavior and uncover patterns related to attention span.

Real-world Scenario

You are working as a Product Data Analyst for a social media platform.

Recently, product teams noticed that average user session duration decreased by **19%**, and content completion rates have started dropping.

Management suspects changes in content behavior but lacks evidence.

Your mission is to perform a statistical investigation and identify factors influencing user attention span.

Objective

Analyze social media user behavior data and identify factors affecting attention duration and engagement.

Use statistical analysis to discover patterns and provide recommendations that improve platform performance.

Dataset Suggestions

Choose one:

- Social media user behavior datasets (Kaggle)
- User engagement datasets
- App session datasets
- Video consumption datasets
- Generate synthetic social media activity data

Tasks to Perform

Phase 1: Data Understanding

- Load dataset
- Understand variables
- Check missing values
- Remove duplicates
- Study data distributions

Questions:

- Total users?
- Average session duration?
- Average engagement level?

Phase 2: Descriptive Statistical Analysis

Calculate:

- Mean
- Median
- Mode
- Variance
- Standard deviation
- Quartiles

Questions:

- What is average attention duration?
- Is session duration normally distributed?

Phase 3: Attention Pattern Investigation

Investigate:

- Content Type vs Session Duration
- Scroll Count vs Attention
- Notifications vs Engagement
- Daily Usage vs Attention Span
- Age Group vs Attention

Visualizations:

- Histograms
- Boxplots
- Heatmaps
- Scatter plots

Questions:

- Which content type retains users longer?
- Do users with excessive scrolling lose attention faster?

Phase 4: Outlier Detection

Detect unusual behavior:

Examples:

- Extremely high usage users
- Very low engagement users
- Users with unusually long sessions

Methods:

- Boxplots
- IQR
- Z-score

Questions:

- Which users show abnormal usage patterns?

Phase 5: Hypothesis Testing

Perform one statistical test.

Example:

Null Hypothesis (H_0): Content type has no impact on user attention span.

Alternative Hypothesis (H_1): Content type significantly impacts attention span.

Possible tests:

- T-Test
- ANOVA
- Chi-square

Interpret results using product/business language.

Phase 6: User Segmentation

Create user categories:

Examples:

- Heavy users
- Short-session users
- Video-first users
- Scroll-heavy users
- High-engagement users

Analyze behavior differences.

Phase 7: Business Insights

Answer:

1. Which factors affect user attention most?
2. Which content types increase engagement?
3. Do notifications improve or reduce attention?
4. Which user groups are losing attention fastest?
5. What product recommendations would you provide?

Expected Deliverables

- ☑ Statistical analysis notebook
- ☑ User behavior report
- ☑ Visualizations
- ☑ Business recommendations
- ☑ GitHub repository
- ☑ README documentation

GitHub Submission Structure

TASK_06_Social_Media_Attention_Span_Study/

|── data/

|── notebook/

|── images/

|── reports/

|── src/

|── README.md

|── requirements.txt

README should include:

1. Problem Statement
2. Dataset Description
3. Statistical Methods
4. User Behavior Findings

5. Visual Insights
6. Recommendations
7. Future Scope

Evaluation Criteria

Criteria	Weight
Problem Understanding	20%
Statistical Analysis	20%
Visualization Quality	20%
Business Insights	20%
Documentation	10%
Innovation	10%